UR5 multi-obstacle transfer and upright motion

Task 4-5 V2 implementation, fresh-seed evaluation and failure diagnosis

The main scene contains three staggered boxes. The variant moves both side pillars inward by 40 mm and raises the right pillar. Both tasks use the same upright endpoints and colliding straight joint interpolation. The PoE model, sphere envelopes and original STOMP stages come from Tasks 1-3. No feasible witness path initializes the optimizer.

Why V1 Task 5 failed

All 20 V1 projected formal paths satisfy orientation (maximum tilt 0.462-0.963 degrees) and the conservative world margin (38.34-69.18 mm). Every path self-collides. Representative mesh checks identify forearm/wrist intersections and, in some runs, base/arm and shoulder/arm intersections. The original objective omits self-collision. Direction projection cannot resolve geometry, and cost-only acceptance discards candidates before feasibility testing. Adjacent V1 seeds also reuse restart streams.

V2 method

A capsule penalty guides separation of non-parent body pairs in both conditions. Public collision visualization supplies the installed meshes. Enclosing capsules can overlap where meshes are separated, so pair targets are capped at the fixed-endpoint capsule gaps. Actual mesh self-collision rejection now operates inside candidate selection. Promising fully feasible proposals take priority even when their cost increases. Task 5 additionally uses the L1 tool-y direction penalty and unchanged damped PoE projection.

Waypoints / rollouts	Iterations / restarts	Self / orientation weight	Budget / noise
61 / 64	200 / 3	60000 / 1000	300 s / 0.55 rad

The planner checks its time target between iterations, including trace writes. A final iteration can finish slightly after 300 s. Statistics retain actual elapsed times.

Acceptance requires fixed endpoints, joint limits, at least 35 mm conservative world clearance, zero actual world and self collisions, and maximum tilt at most 5 degrees for Task 5. Final interpolated checks limit every joint increment to 0.0125 rad. These are discrete checks.

Development uses seeds 5150-5152. Formal seeds 5200-5209 were reserved until freezing. Restart streams use a stride of 100000. Two local workers split the formal seeds by parity. The incomplete capsule-only pilot is preserved separately as abandoned development work.

Fresh formal results

All 50 completed nominal cases enter the statistics: 40 paired scene/task runs and 10 main-scene penalty-only ablations. Runtime summaries include completed failures. Quality summaries use successful runs only.

Scene / mode	Success	Time median (s)	Max tilt (deg)	Gap (mm)
main / Task 4	10/10	30.53	126.55	45.73
main / Task 5	10/10	52.46	0.93	49.17
main / penalty only	0/10	301.48	-	-
variant / Task 4	10/10	29.83	134.52	50.26
variant / Task 5	10/10	48.44	0.95	63.05

Projected Task 5: **20/20** full success, 20/20 meet the upright check and 0/20 fail self-collision checks. V1's projected result remains preserved at 0/20. These are descriptive counts on different fixed seed sets.

Common successful pairs

main: n=10. Median maximum tilt A/B: 126.55/0.93 degrees. Median paired planner time B minus A: 19.03 s.

variant: n=10. Median maximum tilt A/B: 134.52/0.95 degrees. Median paired planner time B minus A: 19.04 s.

Failure labels

main_none: none.

main_projected: none.

main_penalty: proxy_margin=1, orientation=10.

variant_none: none.

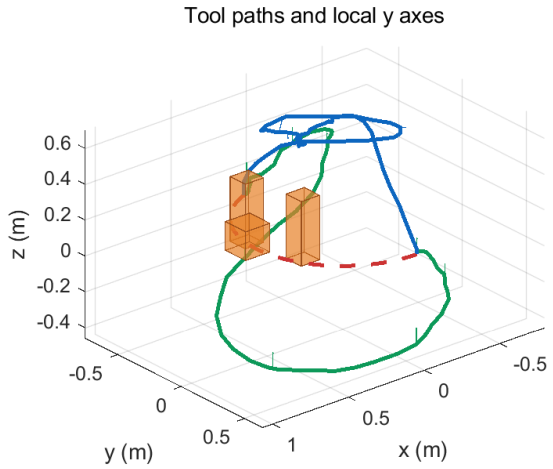
variant_projected: none.

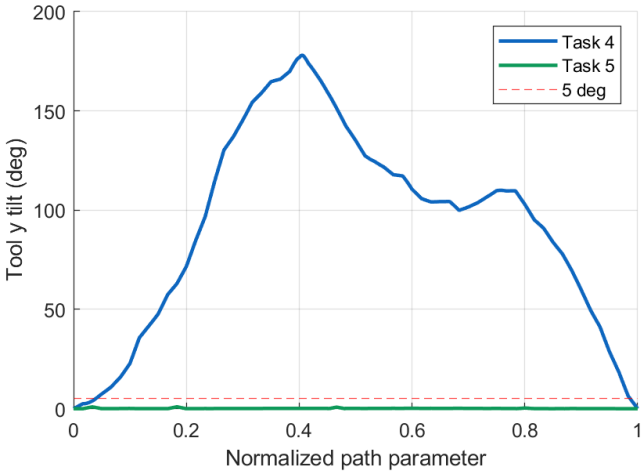
Labels can overlap. Different weighted objective totals are not used to claim better performance. Two workers share this machine, so no cross-version runtime inference or significance claim is made.

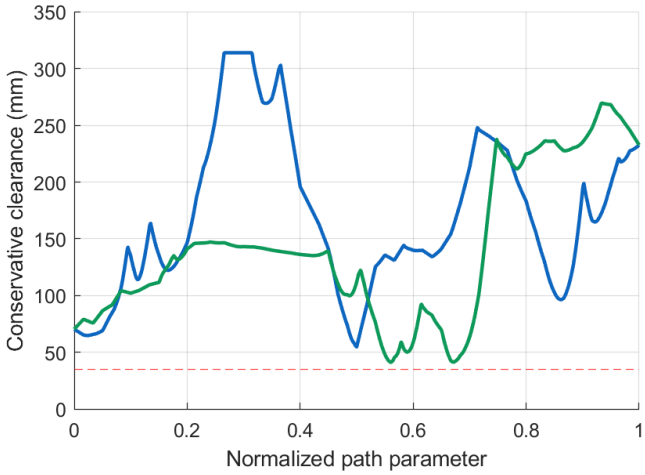
Independent evidence

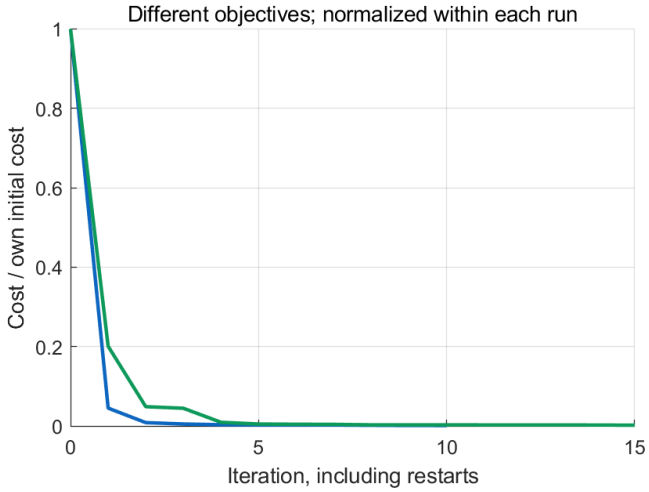
Fresh mesh/clearance validation and complete random-sample/update replay passed for 50 formal runs (1991 checkpoints). The audit also recomputed 256 objective values and checked in-loop safety receipts. Four demonstration runs passed a separate audit. V1's 6458-file manifest still matches byte for byte.

Main scene: paired demonstration







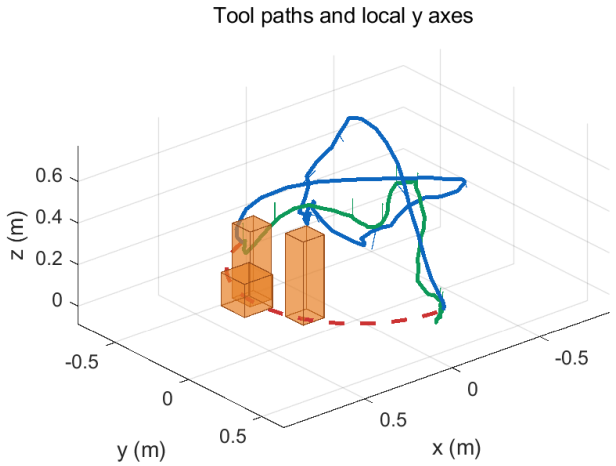


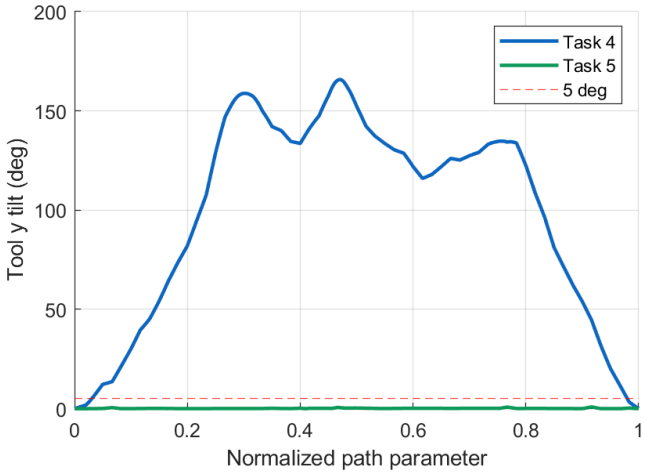
Tool-y arrows show uprightness along normalized path progress. The angular gate is 5 degrees and the conservative clearance gate is 35 mm. Each cost curve is normalized to its own initial objective.

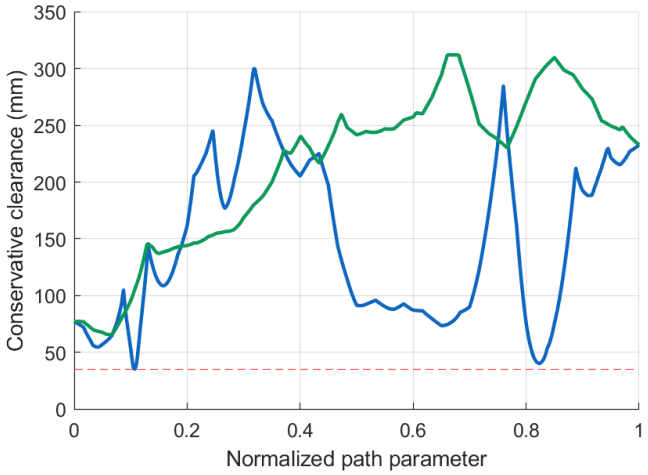
Task 4, seed 5150: minimum conservative clearance 54.67 mm, actual mesh world gap 112.87 mm, maximum tilt 177.995 degrees. All 744 dense states pass world and self checks.

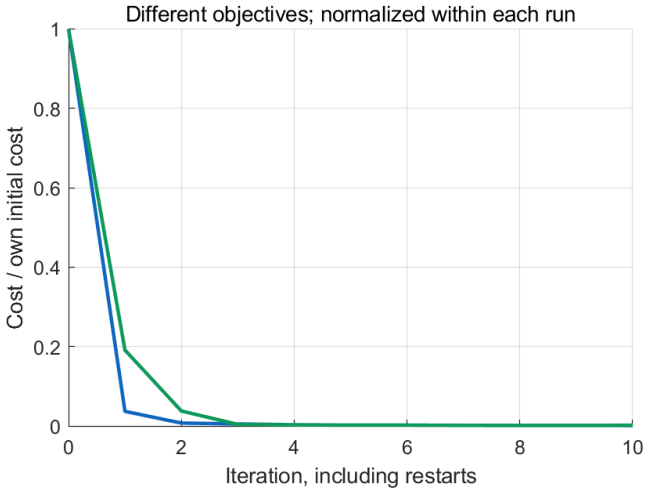
Task 5, seed 5150: minimum conservative clearance 41.14 mm, actual mesh world gap 97.98 mm, maximum tilt 0.956 degrees. All 599 dense states pass world and self checks.

Variant scene: paired demonstration









Tool-y arrows show uprightness along normalized path progress. The angular gate is 5 degrees and the conservative clearance gate is 35 mm. Each cost curve is normalized to its own initial objective.

Task 4, seed 5150: minimum conservative clearance 35.21 mm, actual mesh world gap 88.01 mm, maximum tilt 165.769 degrees. All 639 dense states pass world and self checks.

Task 5, seed 5150: minimum conservative clearance 65.50 mm, actual mesh world gap 146.25 mm, maximum tilt 0.869 degrees. All 617 dense states pass world and self checks.

Contribution and limits

Member C adds the scenes, direction mechanism, self-aware acceptance, independent audits, paired studies and delivery materials. The project still uses a finite seed sample and discrete collision checks. No payload, dynamics or continuous collision guarantee is modeled. Animation progress is not physical time.